Towards a Cartography of Datacentering

A Multimodal Research Agenda

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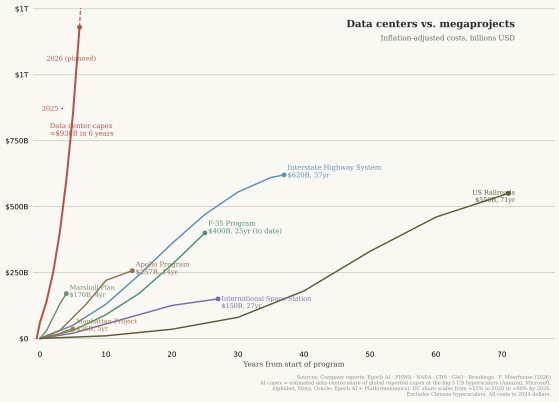
Erasmus University Rotterdam

How Big Is This?



$930B in 6 years of data center capex

- 400 TWh/yr globally
- Millions L/day cooling
- Land: small-town scale
- REIT, PE, SWF capital

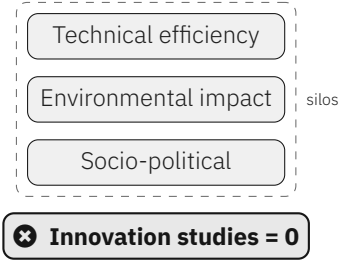
But: private recurring capex ≠ one-off public programme. The comparison does rhetorical work.

4,757 Papers. One Blind Spot.

Technical efficiency	≈ 279
Environmental impact	≈ 114
Socio-political	≈ 15
Innovation studies	**0**

No MLP. No TIS. No SNM.
No sociotechnical regime analysis.

372% growth in 15 years —
and the field has not noticed.



Introducing “Datacentering”

Datacentering = the active, contested, socio-material process of assembling, operating, and reproducing digital infrastructure in place

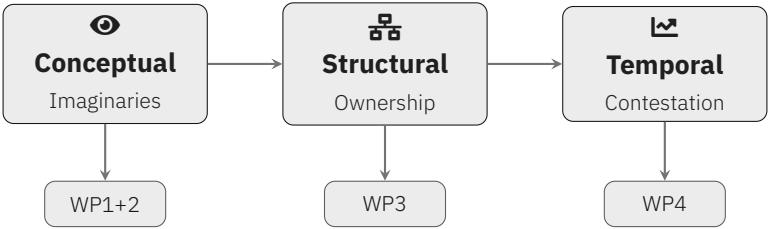
Why a gerund?

- Process, not static object
- *Who* is doing it? *For whom*?
- What is assembled — and foreclosed?

Theoretical grounding

- Orlikowski (2007): sociomaterial entanglement
- Jasanoff & Kim (2015): imaginaries co-produce facts
- “Cloud” *suppresses* what datacentering names

CAS as a Sensemaking Lens



After **Phillips & Ritala (2019)** — sensemaking lens, not a formal model

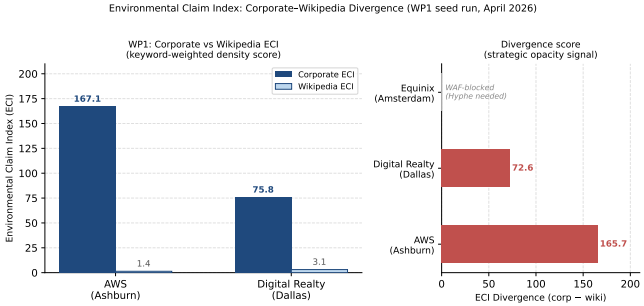
Multi-actor · nonlinear · emergent

A Multimodal Research Agenda

WP	Method	Data	Key output
1	Text NLP	Operator websites; Wikipedia	ECI divergence
2	Satellite + CV	Sentinel-2; Landsat LST; VIIRS	Visual abstraction; thermal footprint
3	Networks	SEC EDGAR; REIT filings; Overture Maps	Ownership graph; urban graph
4	Event detection	GDELT; planning records; social media	Signal-to-response lag

Strategic opacity is a *sociotechnical fact*, not a data gap. A firm that won't publish is still in a REIT filing, a satellite image, and a planning record.

WP1: The Imaginary in Numbers



ECI divergence

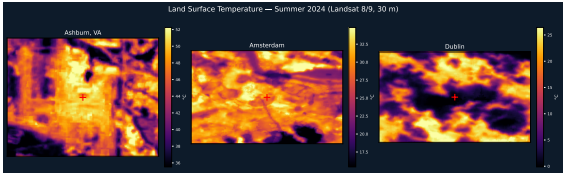
AWS Ashburn: corporate ECI **120**× Wikipedia

Equinix: WAF-blocked the scraper
*Strategic opacity at the **technical** layer*

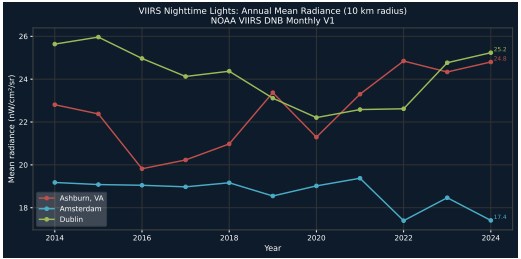
High corporate ECI + low Wikipedia ECI
= placelessness production

The sustainability claim and the territorial fact are decoupled by design.

WP2: What Satellites See



LST thermal maps — Landsat 8/9, summer 2024



Ashburn: **+8.7%** radiance 2014–2024

Only jurisdiction with net growth

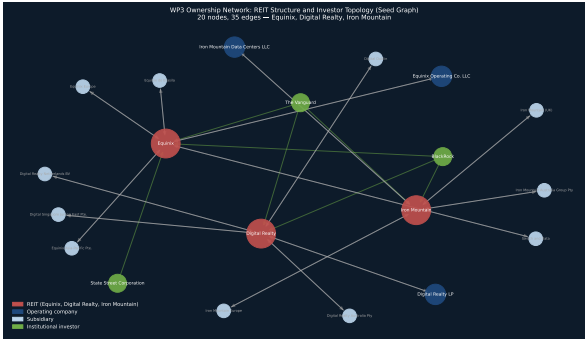
Amsterdam: −9.2%

LED transition — measurement confound

Isolated heat plume in exurban Virginia;

thermal signature invisible to neighbours.

WP3a: Who Actually Owns This?



■ REIT holding ☐ operating ○ subsidiary ● investor

REIT structure

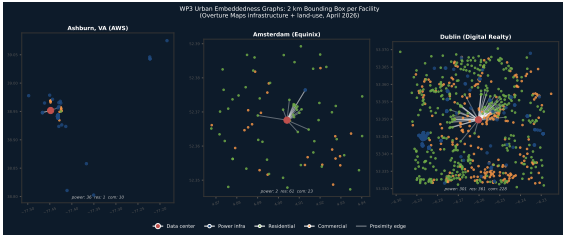
Vanguard, BlackRock, State Street each hold 6–10% across *all three* REITs

No governance counterparty

Operating entity legally separated from asset-owning REIT

When communities contest, there is no single actor that can make a binding commitment.

WP3b: Where It Sits in the City



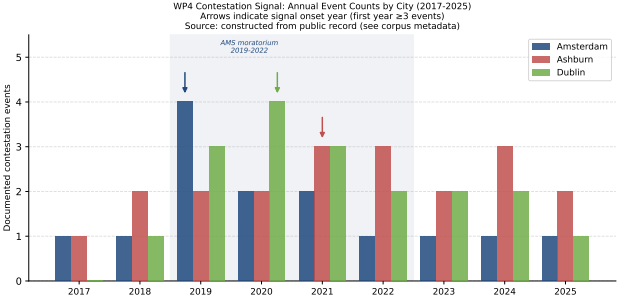
	Ashburn	Amsterdam
Power nodes	36	2
Res. zones	**1**	**61**
Adj. edges	**0**	**16**

Ashburn: dense power grid, **zero residential adjacency**

Amsterdam/Dublin: embedded in urban fabric

Spatial separation is not accidental — it is a governance design choice.

WP4: When Communities Resist



Seed corpus: **58 events**, 4 cities

Dublin: **81% negative-tone**
EirGrid grid cap + An Bord Pleanála

Amsterdam moratorium:
signal onset → response < **1 year**

Ashburn: comparable signal,
zero regime response

The variable is not signal strength.

It is the structure of the receiving system.

Same Signal, Different System

Dimension	N. Virginia	Amsterdam
Ownership	REIT-dominated; dispersed	Mixed; operator-owned, AMS-IX coop
Counterparty	**None**	Municipality + operators
Urban (2 km)	36 power / **1 res.**	2 power / **61 res.**
ECI gap	**+165.7**	Lower; regulatory vocab absorbed
Signal	High	High
Response	**None**	**Moratorium 2019–2022**

Key variable: ownership structure + governance counterparty

Building the Cartography Together

Why open is a theoretical necessity

CAS-like phenomena are too distributed, too opaque for any single team. The commons *is* the method.

What we share

- Facility index (CC BY 4.0)
- NLP / CV / network pipelines (MIT)
- `github.com/babayyy/datacentering-cartography`

Concrete asks

- Local knowledge: disputes, unreported sites
- Company registry or REIT data access
- Methodological critique
- Replication in other jurisdictions

"The cloud is not a metaphor.
It is infrastructure.
And infrastructure can be mapped."